

DRUMMOND ISLAND, MI, USA

WMO: 720375

Lat: 46.007N

Lon: 83.743W

Elev: 203

StdP: 98.91

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 54844

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.3	(c) -20.3	(d) -27.4	(e) 0.3	(f) -23.2	(g) -24.1	(h) 0.4	(i) -19.5	(j) 8.4	(k) -4.2	(l) 7.6	(m) -3.8	(n) 1.1	(o) 340	(p) 0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	27.8	20.3	26.5	19.6	25.0	18.7	22.0	25.8	21.0	24.1	20.2	23.3	2.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.0	24.0	20.0	15.1	22.9	19.0	14.1	21.8	65.5	25.8	61.8	24.0	58.7	23.0	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.4	(b) 6.4	(c) 5.5	(e) -28.3	(f) 30.2	(g) 3.4	(h) 1.9	(i) -30.7	(j) 31.5	(k) -32.7	(l) 32.7	(m) -34.6	(n) 33.7	(o) -37.1	(p) 35.1		
(a) 7.4	(b) 6.4	(c) 5.5	(e) -28.6	(f) 23.5	(g) 3.3	(h) 1.1	(i) -30.9	(j) 24.3	(k) -32.9	(l) 24.9	(m) -34.8	(n) 25.5	(o) -37.2	(p) 26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-8.4	-8.1	-3.1	3.2	10.7	15.8	19.0	18.5	14.8	8.3	2.1	-4.0
	DBStd	10.77	6.74	5.90	6.01	4.35	4.19	3.21	2.93	2.77	3.90	4.15	4.50	5.47
	HDD10.0	2496	569	506	408	206	41	2	0	0	5	85	239	435
	HDD18.3	4687	827	739	666	453	240	89	28	33	118	312	488	693
	CDD10.0	964	0	0	1	3	62	174	280	262	148	32	2	0
	CDD18.3	113	0	0	0	0	3	12	49	37	12	0	0	0
	CDH23.3	725	0	0	0	1	31	97	349	201	45	2	0	0
	CDH26.7	112	0	0	0	0	3	14	67	24	4	0	0	0
(14) Wind	WSAvg	2.3	2.4	2.5	2.4	2.6	2.4	2.0	1.9	1.9	2.0	2.5	2.7	2.6
(15) Precipitation	PrecAvg	1052	84	60	62	81	85	89	85	89	108	117	94	98
	PrecMax	1251	142	115	140	156	134	154	157	148	196	184	153	180
	PrecMin	844	38	21	7	39	42	46	41	48	47	48	39	20
	PrecStd	112	23	23	30	28	25	27	29	26	37	37	29	31
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	5.4	15.0	18.0	27.0	28.5	30.6	28.9	27.3	22.1	15.6	9.1
		MCWB	5.1	3.9	8.7	10.4	18.5	20.1	21.4	21.8	20.7	16.9	12.1	8.6
	2%	DB	3.0	2.5	9.9	15.1	23.7	26.1	28.4	27.2	24.6	18.8	12.1	5.8
		MCWB	2.1	1.6	7.1	9.0	16.2	19.0	20.3	20.6	19.6	15.5	9.8	4.9
	5%	DB	2.3	1.2	6.6	12.5	21.2	24.0	27.1	25.7	22.7	17.1	10.1	4.1
		MCWB	1.5	0.4	3.8	7.0	14.8	17.2	19.7	19.4	18.3	14.0	8.3	3.1
	10%	DB	0.9	0.1	4.1	10.2	18.4	22.3	25.2	24.0	21.3	15.1	8.1	2.8
		MCWB	0.5	-0.6	1.9	5.8	13.0	16.2	19.0	18.6	17.7	12.9	6.8	2.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	4.0	11.7	12.2	19.7	22.2	23.5	23.0	22.0	18.2	12.4	8.8
		MCDB	5.8	5.3	14.6	16.0	24.7	26.6	28.0	27.3	24.8	19.6	14.7	9.1
	2%	WB	2.7	1.8	6.9	10.1	17.7	19.9	22.1	21.9	20.7	16.0	10.2	5.1
		MCDB	3.2	2.4	8.9	13.8	21.9	24.0	26.0	25.6	23.2	18.3	11.7	5.6
	5%	WB	1.4	0.4	4.2	8.0	15.8	18.6	20.9	21.0	19.5	14.5	8.5	3.4
		MCDB	2.0	0.9	6.4	11.3	19.5	22.5	24.5	23.6	21.9	16.5	9.7	4.0
	10%	WB	0.3	-0.6	2.4	6.2	14.1	17.4	20.0	20.1	18.3	13.0	7.2	2.2
		MCDB	0.8	-0.1	3.8	9.3	17.6	20.8	23.8	22.7	20.6	15.0	8.1	2.9
(35) Mean Daily Temperature Range	MDBR		8.0	8.5	9.2	8.8	11.0	10.6	10.4	9.5	9.3	7.5	6.0	6.4
	5% DB	MCDBR	5.4	7.3	10.3	13.6	14.7	13.0	12.3	11.2	10.4	10.2	8.9	6.3
		MCWBR	5.2	6.5	7.3	8.1	8.3	6.6	5.8	5.4	5.9	7.0	6.9	5.9
	5% WB	MCDBR	5.2	7.2	9.7	12.1	12.9	10.8	10.3	9.3	8.9	8.9	8.2	6.1
		MCWBR	5.1	6.8	7.2	8.2	8.0	6.4	5.7	5.4	6.1	6.7	6.8	5.9
(40) Clear-Sky Solar Irradiance	taub		0.290	0.293	0.319	0.354	0.374	0.395	0.394	0.384	0.366	0.331	0.304	0.285
	taud		2.383	2.365	2.370	2.334	2.374	2.349	2.372	2.401	2.439	2.503	2.517	2.457
	Ebn at Noon		812	887	910	903	893	872	867	864	850	835	799	784
	Edh at Noon		72	89	103	118	117	121	117	109	97	78	64	61
(44) All-Sky Solar Radiation	RadAvg		1.18	2.09	3.55	4.60	5.57	6.01	6.01	5.09	3.91	2.31	1.24	0.89
	RadStd		0.14	0.23	0.34	0.46	0.50	0.47	0.28	0.31	0.34	0.25	0.16	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page